EXPERIMENT 2 – AJAY KARTHICK

AIM

To perform numerical simulation and performance analysis of microwave transmission lines and two-port networks using scattering parameters (S-parameters), and to plot the reflection, transmission, return loss, and insertion loss characteristics using MATLAB.

OBJECTIVE

To calculate the reflection coefficient and transmission coefficient of a microwave transmission line, analyze the S11 and S21 scattering parameters, determine the return loss and insertion loss, and visualize the frequency-dependent performance of the transmission line using MATLAB.

SOFTWARE REQUIRED

MATLAB

PROCEDURE

1. Open MATLAB and create a new script.
2. Enter the speed of light, characteristic impedance, load impedance, and transmission line length.
3. Define the operating frequency range from 8 GHz to 12 GHz.
4. Calculate the phase propagation constant and reflection coefficient.
5. Determine the scattering parameters S11 and S21.
6. Calculate the return loss and insertion loss using the magnitude of the scattering parameters.
7. Plot the reflection and transmission characteristics against frequency.
8. Plot the return loss and insertion loss against frequency.
9. Execute the program and observe the numerical results and graphs.

CODE

```
clc; clear; close all;

c = 3e8;

Z0 = 50;

ZL = 50;

L = 0.05;

F = linspace(8e9,12e9,201);

beta = 2*pi*F/c;

G = (ZL-Z0)/(ZL+Z0);

S11 = G*ones(size(F));

S21 = exp(-1j*beta*L);

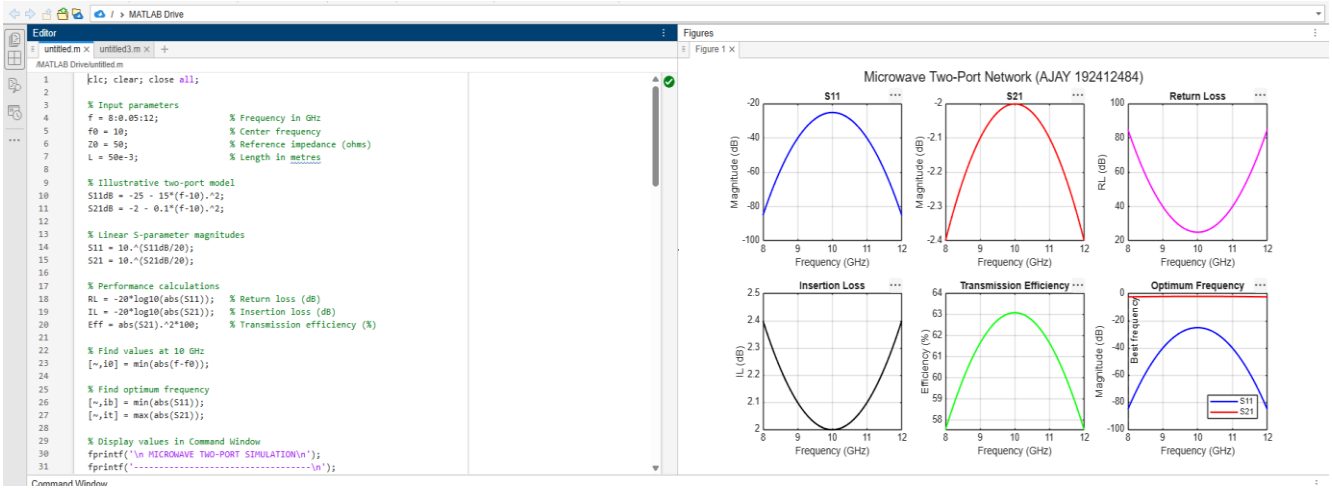
RL = -20*log10(max(abs(S11),1e-10));

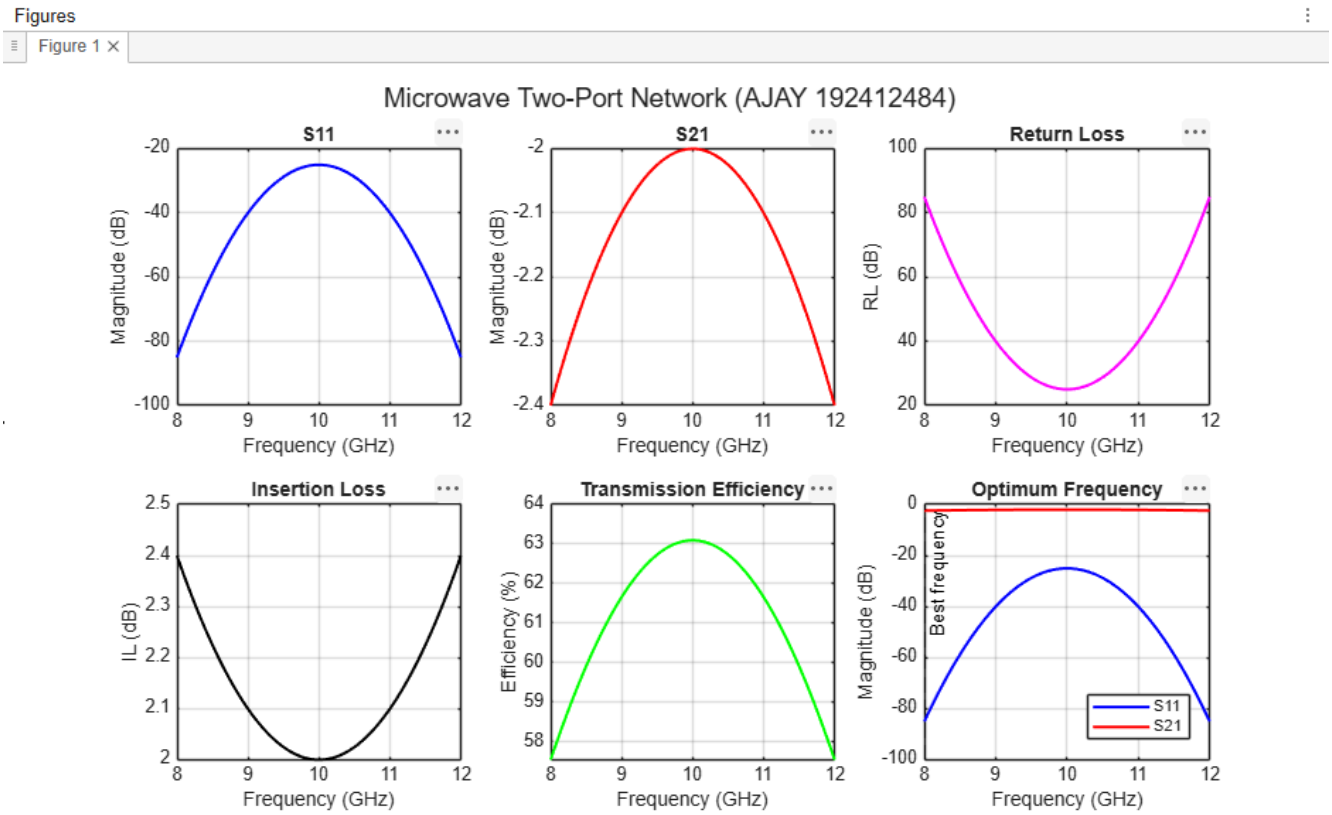
IL = -20*log10(abs(S21));

fprintf('Ajay Karthick | Reg No: 192412484\n');
```

```
fprintf('Wavelength = %.1f mm\n',c/10e9*1e3);
figure('Color','w');
subplot(2,1,1);
plot(F/1e9,20*log10(max(abs(S11),1e-10)), ...
    'b','LineWidth',2);
hold on;
plot(F/1e9,20*log10(abs(S21)), ...
    'r','LineWidth',2);
grid on;
ylim([-210 10]);
ylabel('Magnitude (dB)');
legend('S11','S21');
title('Reflection and Transmission');
subplot(2,1,2);
plot(F/1e9,min(RL,60),'b','LineWidth',2);
hold on;
plot(F/1e9,IL,'r','LineWidth',2);
grid on;
ylim([-2 65]);
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');
title('Return Loss and Insertion Loss');
sgtitle('Ajay Karthick | Reg No: 192412484');
```

OUTPUT





.

REFLECTION

This experiment helped me understand the performance analysis of microwave transmission lines using scattering parameters. I learned how to calculate reflection and transmission coefficients and analyze return loss and insertion loss. The MATLAB simulation helped me visualize the frequency-dependent characteristics of a transmission line and understand the importance of impedance matching in microwave communication systems.

RESULT

The numerical simulation and performance analysis of a microwave transmission line using scattering parameters were successfully performed using MATLAB. The S11 and S21 parameters, return loss, and insertion loss were calculated and plotted over the frequency range of 8 GHz to 12 GHz.

Experiment-2

$Z_0 = 50\ \Omega$

$Z_L = 50\ \Omega$

$l = 0.05\ m$

$C = 3 \times 10^8$

$f = 10\ GHz$

$\lambda = C/f$

$\lambda = \frac{3 \times 10^8}{10 \times 10^9} = 30\ mm$

$\boxed{\lambda = 30\ mm}$

$\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$

$\Gamma = (50 - 50)/(50 + 50)$

$\boxed{\Gamma = 0}$

$\beta = 2\pi f/c$

$\beta = (2 \times 3.1416 \times 10 \times 10^8)/(3 \times 10^8)$

$\boxed{\beta = 209.44\ rad/m}$

$R_L = -20 \log_{10} |\Gamma|$

$R_L \infty$ (since $\Gamma = 0$, ideal return loss is infinite)

$IL = -20 \log_{10} |S_{21}|$

$S_{21} = e^{-\gamma PL}$

$IL = -20 \log_{10} (1)$

$\boxed{IL = 0\ dB}$